LoRa Push-to-Talk Walkie Talkie

Project Proposal

Team Members: S A Aravind Eswar (EE24BTECH11053)
Siddhanth Y (EE24BTECH11059)

Date: March 2026

Application / Problem Statement

Conventional walkie-talkies operate over licensed FM/VHF/UHF bands, which are subject to regulatory constraints, interference from other users, and limited range in urban environments. They are also typically bulky and expensive. At the same time, consumer-grade Bluetooth and Wi-Fi radios offer short range and require existing infrastructure.

There exists a clear need for a **low-cost, long-range, license-free voice communication device** suitable for scenarios such as:

- Outdoor activities (hiking, camping, field operations) where cellular coverage is unavailable.
- Campus or community communications without reliance on mobile networks.
- Emergency or off-grid communication links requiring robust, low-power operation.

The proposed device targets ranges of **2–5 km in urban areas and up to 15 km in open terrain**, operating entirely within the **licence-exempt 865–868 MHz band** in India.

Proposed Solution

The solution is a self-contained, push-to-talk (PTT) walkie-talkie based on the **LoRa (Long Range) radio protocol**, with onboard speech compression via **Codec 2**. Two identical units will be built, each capable of transmitting and receiving voice.

System Architecture

The signal chain for transmission is:

Microphone → ESP32 (Codec 2 Encode) → LoRa Module (TX) → Antenna

And for reception:

Antenna → LoRa Module (RX) → ESP32 (Codec 2 Decode) → Class-D Amplifier → Speaker

Key Technology Choices

LoRa Radio – RYLR998 Module

LoRa (Long Range) is a low-power wide-area modulation scheme well suited to battery-operated embedded devices. The RYLR998 operates in the 868/915 MHz band, supports output power up to +22 dBm, and achieves a receive sensitivity of −129 dBm, enabling multi-kilometre links with very modest transmit power. At the Codec 2 bitrates used (450–3200 bps), the available LoRa air-data rate is more than sufficient.

Microcontroller – ESP32-WROOM-32
The ESP32 is a dual-core 240 MHz processor with adequate computational headroom to run real-time Codec 2 encoding and decoding. It communicates with the LoRa module over SPI and with the I2S microphone directly over the I2S bus, eliminating the need for an external ADC.

Speech Codec – Codec 2
Codec 2 is an open-source, low-bitrate speech codec specifically designed for voice over narrow-band radio links. At 1200 bps it produces intelligible speech; at 3200 bps the quality is comparable to FM radio. This drastically reduces the amount of data that must be transmitted over the LoRa link.

Audio Front-End
An I2S MEMS microphone provides digital audio directly to the ESP32 with a 60–65 dB SNR and no analogue amplification stage. On the output side, a PAM8403 Class-D amplifier (up to 3 W into 4 Ω, ∼85% efficient) drives an 8 Ω speaker.

Power System
The PCB design implements a lithium-ion battery management and protection circuit (BQ21040 charger, BQ29700 protection) alongside two TPS6300x buck-boost converters, producing regulated 5 V and 3.3 V rails from the battery. USB-C (USB4085-GF-A receptacle) is used for charging. A CP2102 USB-to-UART bridge enables firmware flashing.

Communication Regulatory Compliance

The device operates in the 865–868 MHz licence-exempt band as defined by WPC India, with transmit power limited to ≤14 dBm to meet CE and regional regulatory requirements, well within the permitted 1 W (30 dBm) envelope.

Practical Feasibility

Component Availability

All components are available through domestic Indian suppliers (primarily robu.in), ensuring short lead times and no import complications. The ESP32-WROOM-32 and PAM8403 are commodity parts stocked by multiple distributors. The RYLR998 LoRa module is a purpose-built transceiver with an AT-command interface, significantly reducing RF development effort.

Cost Effectiveness

The total prototype cost of approximately **INR 5000 for one units** compares favourably against commercial walkie-talkies with similar range, which typically retail at INR 3,000–8,000 per unit without the flexibility of open-source firmware or licence-exempt operation.

At scale, the BOM cost would reduce significantly given the commodity nature of most components.

Technical Risk Assessment

Risk	Mitigation	Severity
LoRa bandwidth insufficient for voice	Codec 2 compresses voice to <3.2 kbps; LoRa supports up to 30 kbps	Low
ESP32 too slow for real-time Codec 2	Codec 2 has verified ESP32 ports at 1200–3200 bps	Low
RF regulatory non-compliance	Limit TX power to ≤14 dBm via AT command	Low
Audio latency affecting PTT experience	Codec 2 frame size is 40 ms; acceptable for PTT use	Medium

Table 1: Risk Assessment

Project Timeline

Weeks 1–2: Component procurement and environment setup.
Weeks 3–7: PCB assembly, firmware development, and first prototype testing.
Weeks 8–12: Range testing, design iteration, and final product build. Documentation maintained throughout.

Competitive Analysis

The table below benchmarks the proposed device against representative products across three market tiers: consumer/budget, mid-range professional, and open-source LoRa alternatives.

Analysis by Tier

Budget tier (INR 2,000–4,000 per pair) – Baofeng UV-5R, Retevis RT22. The Baofeng UV-5R is the dominant budget option globally: inexpensive and capable, but it operates on analogue FM and requires an amateur radio licence to transmit legally in India, making it unsuitable for unlicensed use. The Retevis RT22 is licence-free but limited to short-range analogue UHF with no digital compression, resulting in lower effective range and no path for firmware customisation. Our device enters this tier in cost while significantly outperforming both on range.

Mid-range professional tier (INR 10,000–25,000 per pair) – Motorola XT185. The Motorola XT185 is a polished, IP54-rated PMR446 business radio with up to 24 hours of battery life and a claimed 8 km open-field range. However, in urban conditions its practical range drops to 1–2 km, it is purely analogue, and the pair retails at roughly

Product	Technology	Range (Urban / Open)	Licence Required	Open Source	Cost (pair, INR)
Our Device (Loki)	LoRa + Codec 2	2–5 km / 15 km	No (865–868 MHz exempt)	Yes	∼5,600
Baofeng UV-5R	Analogue FM VHF/UHF	1–3 km / 5–8 km	Yes (Ham licence)	No	∼2,000
Motorola XT185	Analogue PMR446	1–2 km / 8 km	No	No	∼16,000
Retevis RT22	Analogue UHF	1–3 km / 5 km	No (licence-free UHF)	No	∼3,000
LILYGO T3-S3 MVRS	LoRa + Codec 2 (dev kit)	2–5 km / 10+ km	No	Yes	∼4,500
Motorola MOTOTRBO R2	Digital DMR UHF/VHF	3–5 km / 10 km	Yes (business licence)	No	∼30,000+

Table 2: Competitive Landscape – representative products across price tiers

INR 16,000–20,000 in India – nearly three times our prototype cost – with no provision for firmware updates or feature extension.

Enterprise tier (INR 30,000+ per pair) – Motorola MOTOTRBO R2. DMR digital radios such as the MOTOTRBO R2 offer superior audio clarity and longer range but mandate a licensed business-band frequency assignment, involve significant recurring regulatory overhead, and are priced far beyond any hobbyist or low-budget deployment scenario.

Open-source LoRa tier – LILYGO T3-S3 MVRS. The LILYGO T3-S3 MVRS is the closest technical analogue to our device: it also uses ESP32 + LoRa + Codec 2 on the same FSK/LoRa modulation at 914.6 MHz. At approximately $27 (INR 2,250) per bare board, two units cost roughly INR 4,500, but the kit ships as a bare development board with no enclosure, no battery management, no charging circuitry, and no push-to-talk hardware – requiring significant additional integration work. Our design integrates all of these elements on a custom PCB.

Key Differentiators

Feature	Our Device	Baofeng UV-5R	Motorola XT185	LILYGO T3-S3	MOTOTRBO R2
Licence-free operation (India)	✓	×	✓	✓	×
Urban range >2 km	✓	∼	×	✓	✓
Open-source firmware	✓	×	×	✓	×
Integrated battery + USB-C charging	✓	×	✓	×	✓
Digital voice (noise resilient)	✓	×	×	✓	✓
Cost per pair < INR 8,000	✓	✓	×	✓	×
Production-ready enclosure	–	✓	✓	×	✓

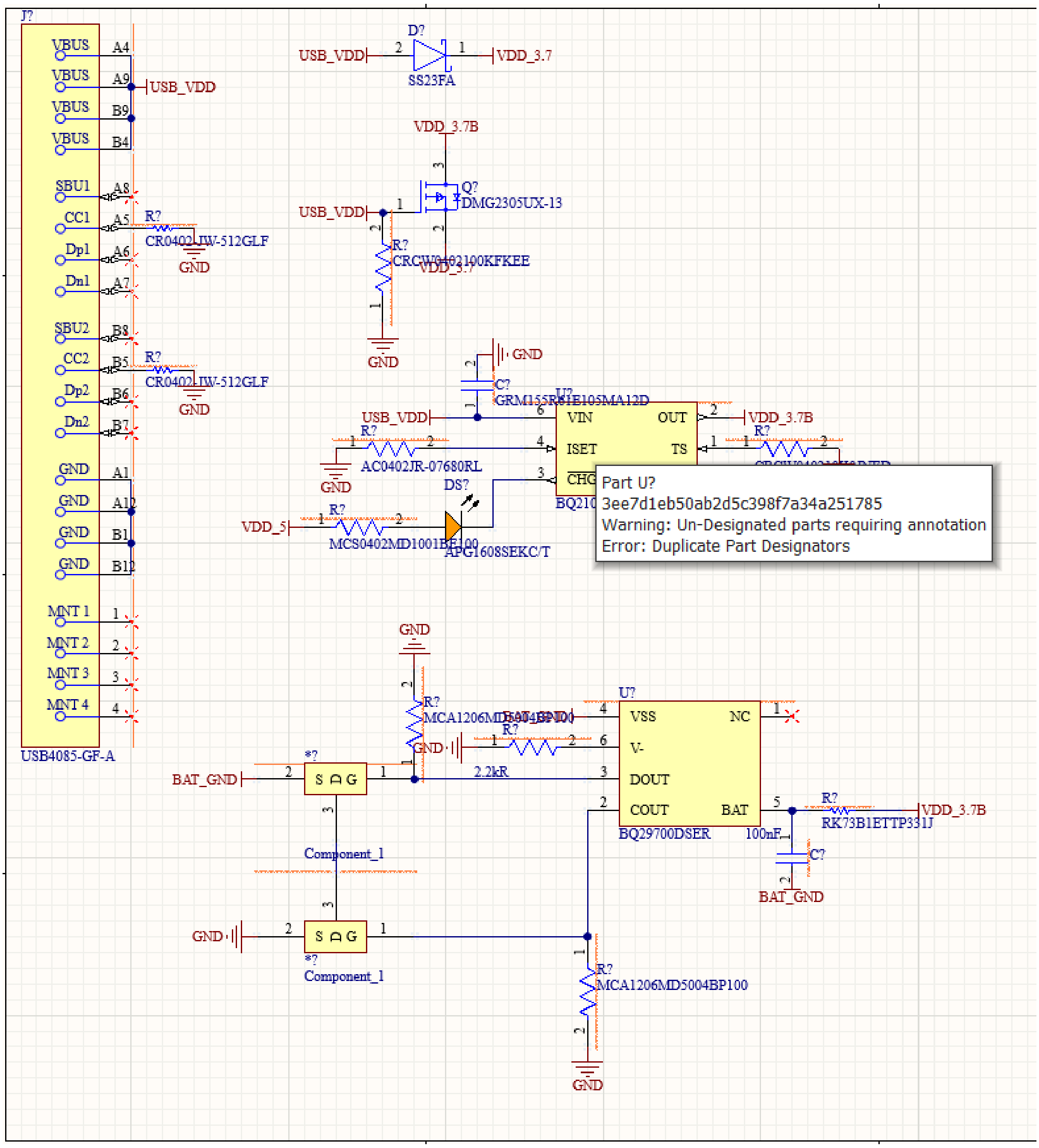
Table 3: Feature comparison matrix (✓ = yes, × = no, – = in progress)

In summary, the proposed Loki device occupies a currently vacant position in the market: a **licence-free, digitally compressed, long-range, open-source walkie-talkie** at a sub-INR 3,000 per-unit cost. No commercially available product combines all four of these attributes simultaneously at this price point.

Breakdown

Charging

Features current surge protection, smart power switching when charging.



Uses BQ21040 for charging and BQ29700 for surge protection.

Buck-Boost Converters

Uses Buck-Boost converters for stable constant voltage output.

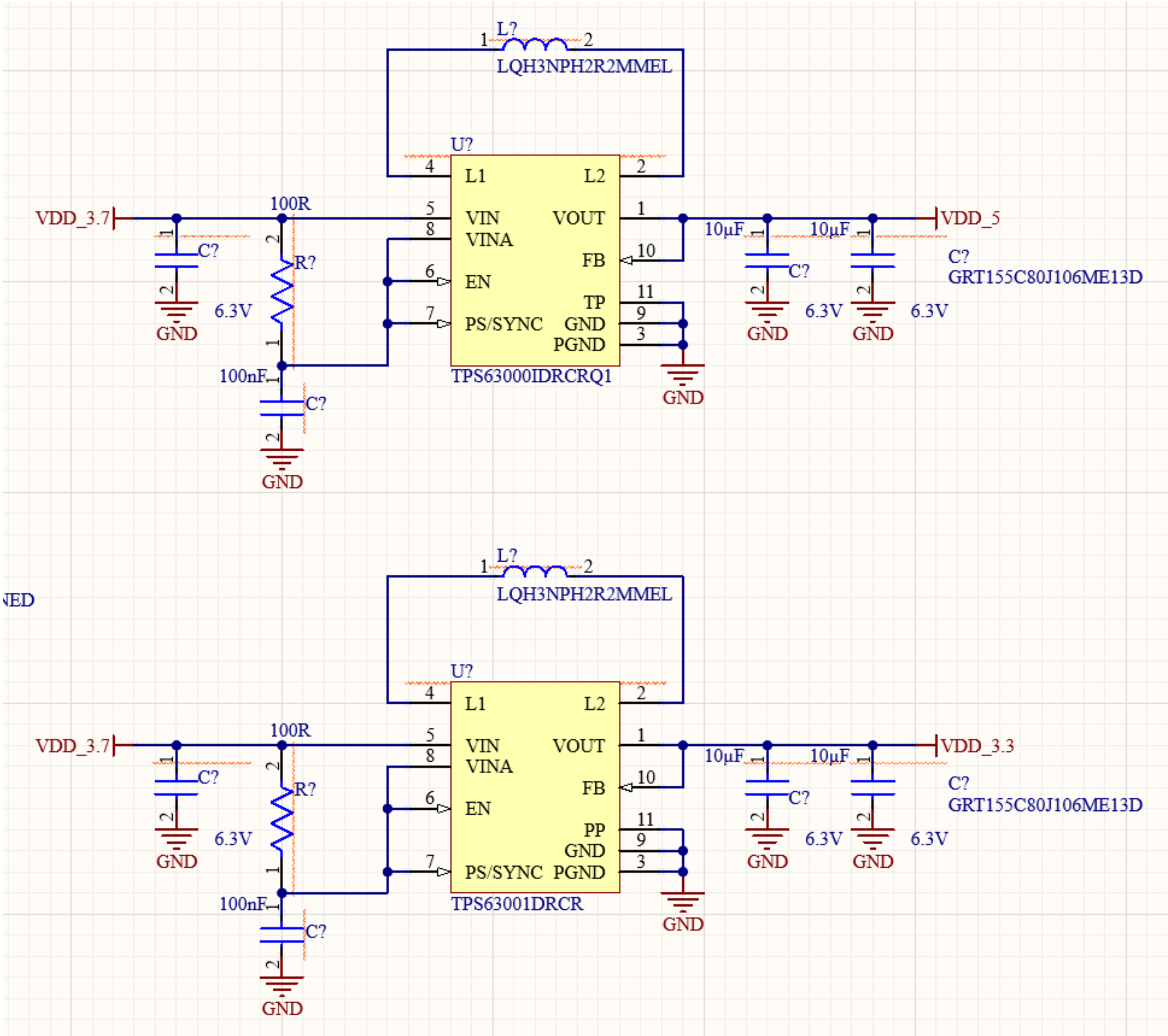


Figure 1: $5V$ and $3.3V$ Buck-Boost converters (top and bottom respectively)

Uses TPS6300X series IC for operation.

Conclusion

The LoRa PTT walkie-talkie is a technically sound and cost-effective solution to the problem of long-range, infrastructure-independent voice communication. The combination of LoRa's propagation characteristics, Codec 2's efficiency, and the ESP32's processing capability creates a feasible and practical device that can be built entirely from commercially available, domestically sourced components within a modest budget.